

Herbivory, predation, disease

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Learning objectives

1. Understand the role of natural enemies for maintaining diversity
2. Explain the relationship between growth and defense and know common defense strategies
3. Model an herbivore attacking adults and seeds

last lecture: maintaining diversity, density dependence lecture mentioned natural enemies

Natural enemies - remove this section?

- high density - more susceptible to disease, attract herbivores and predators
- differentiation leads to diversity *species with different natural enemies coexist because more self-limited than limited by other species*
- Janzen-Connell hypothesis: host-specific enemies reduce seedling survival in dense areas
- apparency theory: aggregation is more easily detected by enemies, vulnerable to herbivores
- enemy release hypothesis: invasive plants successful because enemies aren't present

Natural enemies in different life cycles

create table and add 1 example then ask for other examples

- seed in seed bank
 - seed predators - soil invertebrates & rodents
 - pathogens
- seedling
 - insect herbivores - beetles, aphids
 - pathogens
- reproductive adults
 - insect herbivores
 - grazing mammals
 - pathogens

Plant defense against natural enemies

life history theory - talked about tradeoffs plants make between growth, stasis, reproduction because of limited energy

same with defense, allocate energy to maintenance functions, growth or defense

- growth-defense tradeoff
 - grow fast: rapid germination, low defense, higher herbivory *or predation or disease*
 - well-protected: slow growth, high defense, lower herbivory

trading off investment of energy

- dispersal-defense tradeoff

- diffuse seed dispersal: hard to locate, fewer defenses
- seed defense: aggregated seeds near parent, more defenses
- attract animals *for dispersal*, deter predators *in species that require animal seed dispersers*

Defense strategies

- physical: leaf toughness, spines, latex, hard seeds
- chemical: secondary metabolites *bitter, spicy, etc*, delayed ripening, masting *local satiation*

Modeling natural enemies

modeling adult herbivory

- model plant & herbivore pops
- plant populations:
 - plants: Beverton-Holt model *just density dependence*

$$N_{t+1} = \frac{\lambda N_t}{1 + \alpha N_t}$$

now we need to add the fact that herbivores are eating adult plants and lowering the population size

- a = attack rate *rate that a single herbivore finds and consumes plants*
- H = herbivore abundance

$$N_{t+1} = \frac{\lambda N_t}{1 + \alpha N_t} - a N_t H_t$$

- herbivore populations:
 - b = conversion efficiency *rate that consumed plants converted into new herbivores*
 - m = herbivore mortality

$$H_{t+1} = b(a N_t H_t) - m H_t$$

herbivore population determined by plant consumption minus mortality

modeling seed predator

this is how you'll do it in the lab, model with seed survival, germination rate, fecundity, rather than lambda

$$P_t = \frac{f g N_t}{1 + \alpha g N_t}$$

- seed consumption:
 - C_t - seeds/plants consumed
 - a - attack rate
 - h - handling time *rate that herbivore finds and processes a plant before next plant*
 - H_t - herbivore population

$$C_t = \frac{a P_t H_t}{1 + a h P_t}$$

Full plant model:

$$N_{t+1} = s(1 - g)N_t + P_t - C_t$$

- seedbank + new seeds - new seeds consumed
- herbivore population:
 - s_H
 - b

$$H_{t+1} = s_H H_t + bC_t$$